

Light: Unit Review

Year 9 Science · end-of-year exam

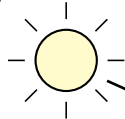
1. Seeing light: sources, materials, shadows
2. Colour: mixing light, objects, filters
3. Reflection: plane mirrors
4. Refraction and dispersion
5. The eye and lenses

Do Now

- 1 Is the Moon a light source or a light reflector?
- 2 Is baking paper transparent, translucent or opaque?
- 3 What colour does a green leaf look in red light?
- 4 A ray hits a plane mirror with an angle of incidence of 35° . What is the angle of reflection?
- 5 Light passes from air into glass at an angle. Does it bend towards or away from the normal?

The one idea

light source



eye



object

Light travels in **straight lines**.

When it meets something it is:

- reflected
- absorbed
- transmitted
- refracted

Sources and reflectors

Light source: emits its own light

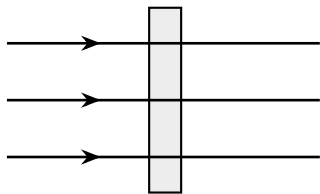
Sun, candle flame, glow-worm, phone screen

Light reflector: only reflects light

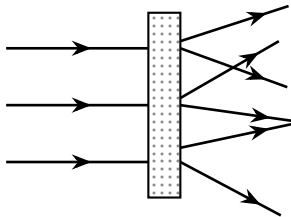
Moon, planet, mirror, this page

We see an object when light from it **enters our eyes**.

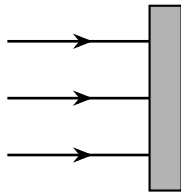
Transparent, translucent, opaque



transparent



translucent



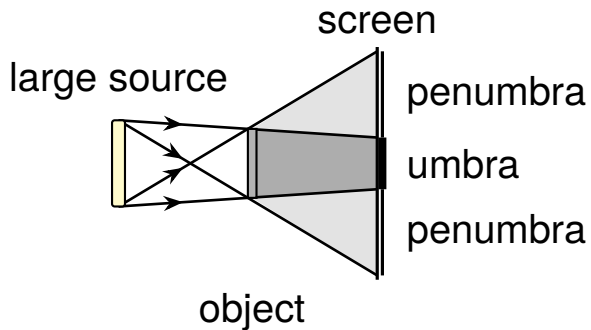
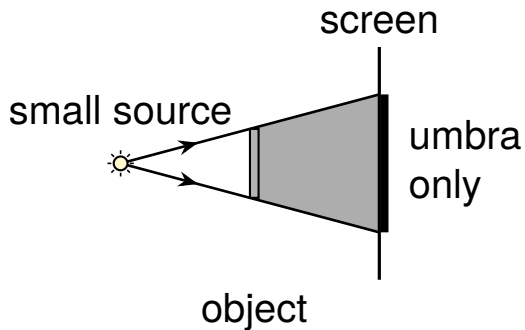
opaque

Three properties of light

1. Travels in **straight lines**
2. Travels very fast: 300 000 km/s
3. Travels through a **vacuum** (sound cannot)

Evidence for straight lines: sharp shadows, a laser beam in smoke, a candle seen only through a straight tube.

Shadows

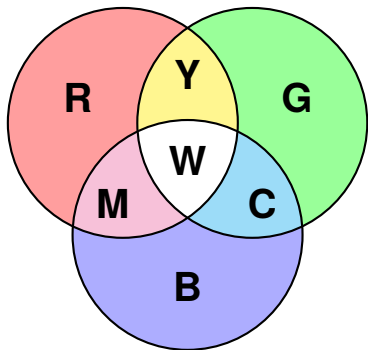


Practice

- 1** Light from the Sun takes about 500 seconds to reach the Earth. Calculate the distance from the Sun to the Earth in km.
- 2** A ball is held between a small lamp and a wall. The ball is moved closer to the wall. What happens to the size of its shadow? Explain.
- 3** In a storm you see the lightning before you hear the thunder. Explain why.

Primary colours of light

White light = red, orange, yellow, green, blue, indigo, violet



R red, **G** green, **B** blue

Y yellow = $R + G$

M magenta = $R + B$

C cyan = $G + B$

W white = $R + G + B$

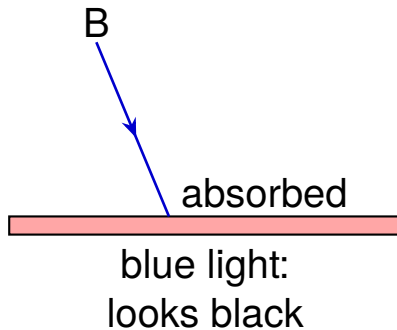
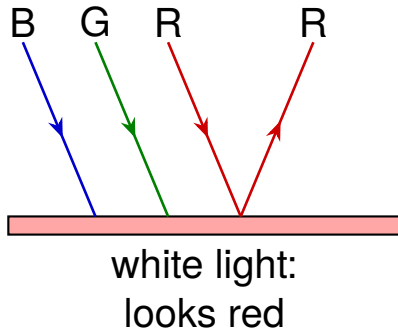
Practice

Predict

A red apple is lit with blue light in a dark room.

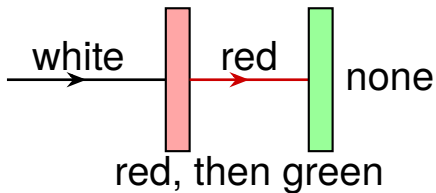
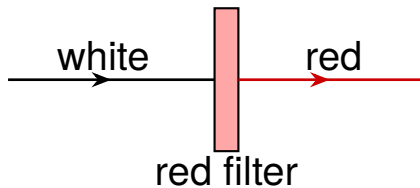
What colour does the apple look?

Coloured objects



An object **reflects** its own colour and **absorbs** the rest.

Filters



A filter **transmits** its own colour and **absorbs** the rest.

Practice

- 1 Red and green spotlights overlap on a white wall. What colour is the overlap? A blue spotlight is then added to the overlap. What colour is it now?
- 2 A white shirt with blue stripes is lit with red light. Describe what you see.
- 3 White light passes through a blue filter and then a red filter. What comes out? Explain.

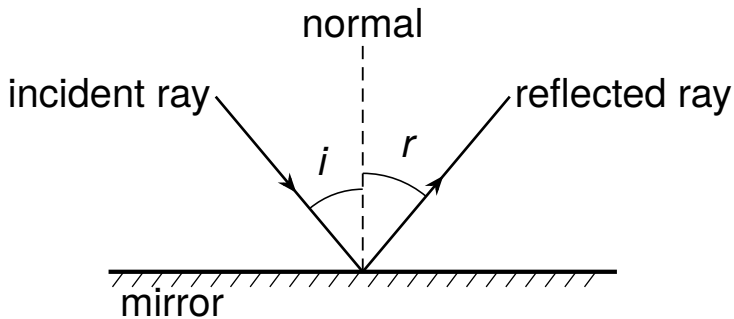
Workbook · In-class Part A

Part A: Shadows and colour

Work on the in-class sheet.

About 7 minutes. Use a ruler for every ray.

The law of reflection



angle of incidence = angle of reflection

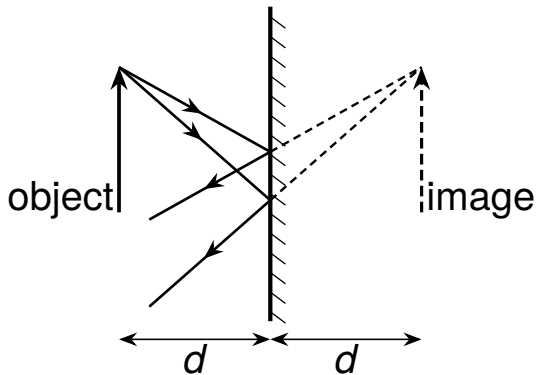
Both are measured from the **normal**.

Practice

Predict

A ray of light hits a plane mirror at 15° to the mirror surface.
What is the angle of reflection?

The image in a plane mirror



- same size
- same distance behind
- upright
- laterally inverted
- virtual

Real or virtual?

	Real image	Virtual image
Light rays	actually meet	only appear to come from it
On a screen?	yes	no
Example	projector, retina	plane mirror

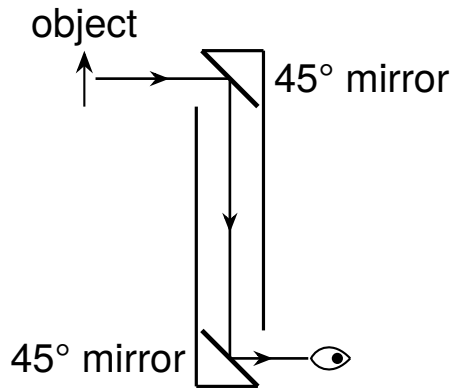
on the van

AMBULANCE

in a mirror

AMBULANCE

Periscope



Two **parallel** mirrors at **45°**

Light turns through 90° at each mirror

See over or around obstacles

Practice

- 1 A student stands 1.5 m in front of a plane mirror. How far is the student from their image?
- 2 Why is AMBULANCE written back-to-front on the front of an ambulance?
- 3 Can the image in a plane mirror be formed on a screen? Explain.

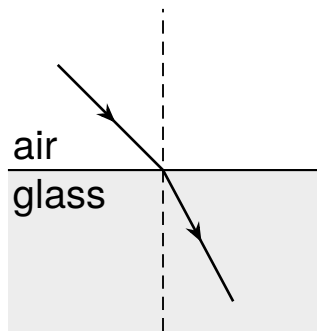
Workbook · In-class Part B

Part B: Reflection

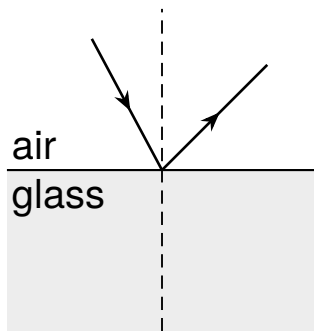
Work on the in-class sheet.

About 7 minutes. Ruler and protractor.

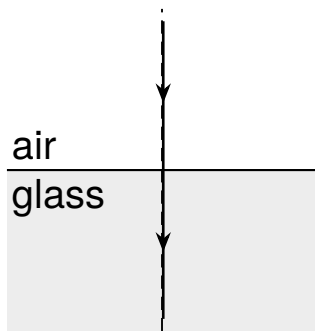
Refraction: why light bends



towards normal



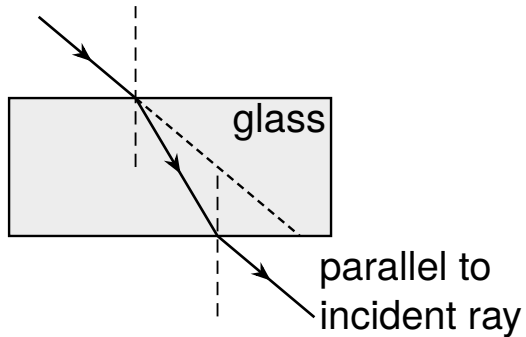
away from normal



no bending

Light **changes speed**. Slower $\rightarrow$ **Towards the Normal**.

Glass blocks

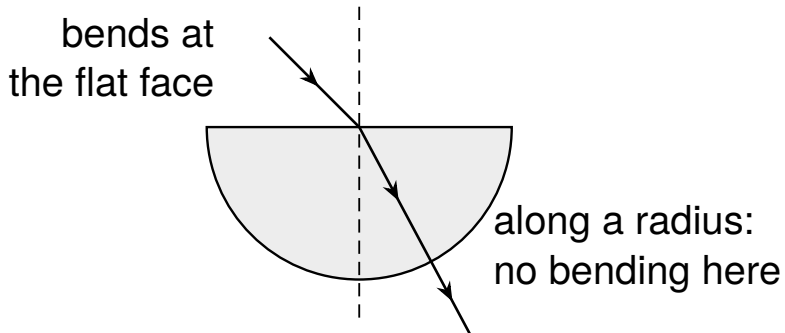


In: towards the normal

Out: away from the normal

Leaves **parallel**,
shifted sideways

Semicircular block



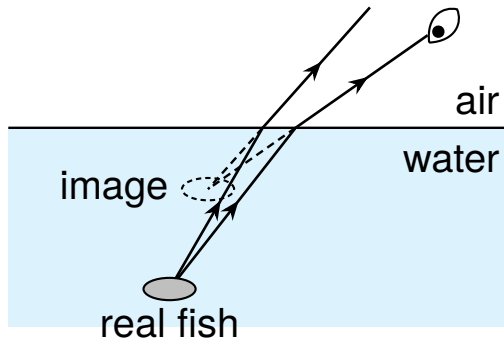
Practice

Predict

A spear fisher sees a fish under the water.

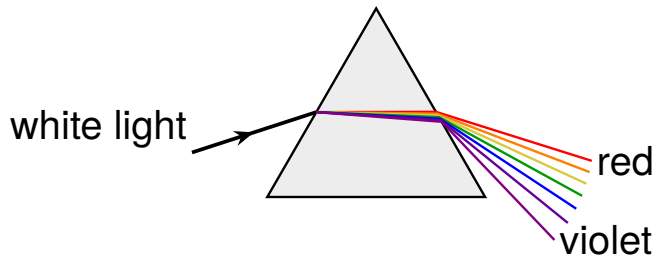
Should they aim at the fish, above it, or below it?

Things under water look closer



1. water $\rightarrow$ air: speeds up
2. bends away from the normal
3. brain traces rays back in straight lines
4. image is higher: virtual

Dispersion



Each colour travels at
a different speed

Violet bends **most**

Red bends **least**

Beyond Year 9: refractive index

$$n = \frac{c}{v} = \frac{\text{speed in a vacuum}}{\text{speed in the material}}$$

water 1.33

glass 1.5

diamond 2.42

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Larger n : slower light, bends more towards the normal.

Practice

- 1 A ray of light enters a glass block along the normal. What happens to it?
- 2 A coin lies in an empty cup, just hidden by the rim. When water is poured in, the coin comes into view. Explain why.
- 3 Which colour is refracted most by a prism, and which least?

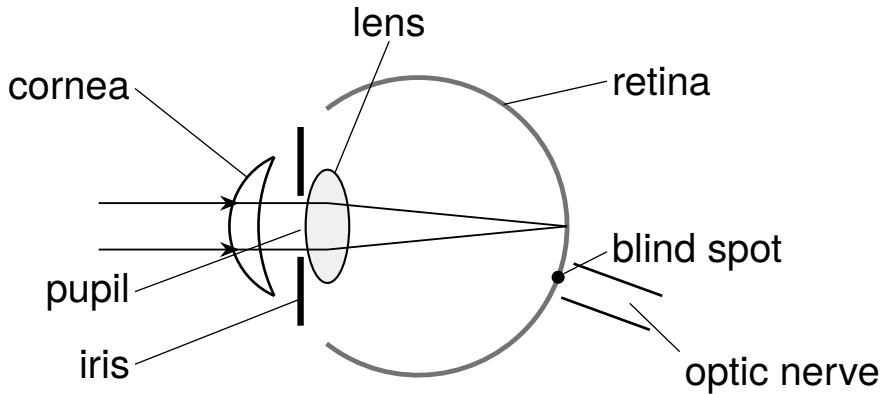
Workbook · In-class Part C

Part C: Refraction

Work on the in-class sheet.

About 6 minutes.

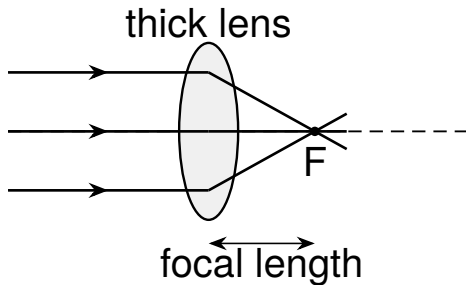
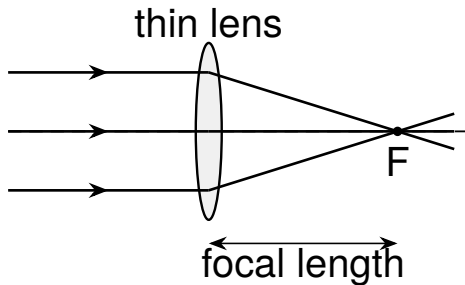
The eye



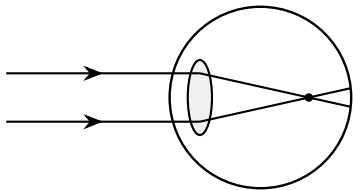
Pupil, image, blind spot

- Dim light: the **iris** makes the pupil **bigger**
- Bright light: the pupil gets **smaller**
- Image on the retina: **real** and upside down
- Blind spot: no light-sensitive cells

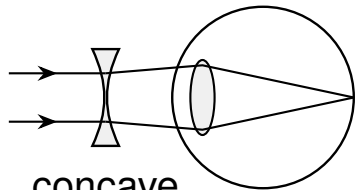
Convex lens: focal point



Short sight



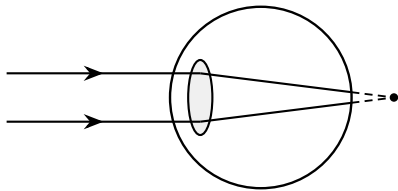
focus in front
of the retina



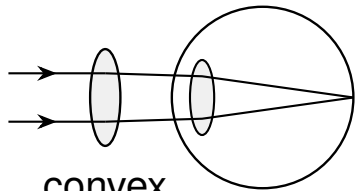
concave
lens

focus on
the retina

Long sight



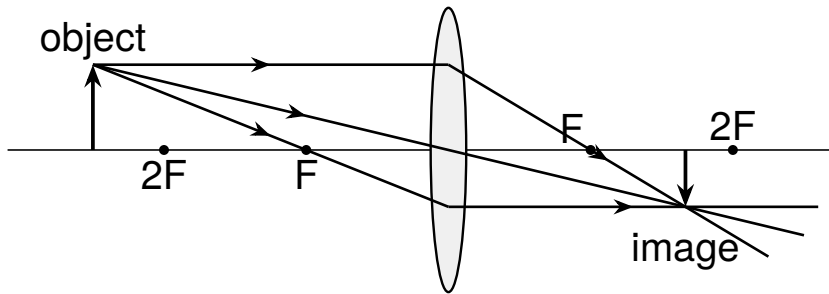
focus behind
the retina



convex
lens

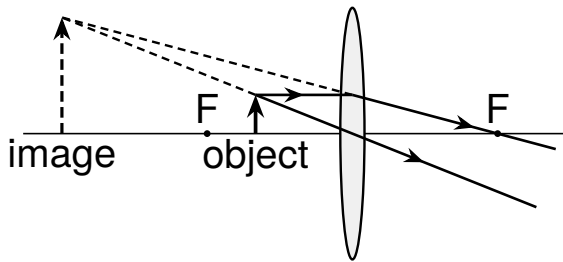
focus on
the retina

Beyond Year 9: images from lenses

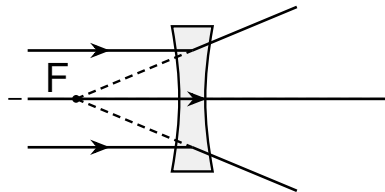


Object beyond $2F$: image real, inverted, smaller

Beyond Year 9: magnifier and concave lens



Inside F: virtual, upright, bigger



Concave: virtual,
upright, smaller

Practice

- 1 What happens to the size of your pupil when you walk into a dark room? Which part of the eye causes this?
- 2 A student can read a book clearly, but the whiteboard looks blurred. Name the condition and the type of lens that corrects it.
- 3 A thin and a thick convex lens are made of the same glass. Which one has the shorter focal length?

Workbook · In-class Part D

Part D: The eye and lenses

Work on the in-class sheet.

About 5 minutes.

Ways to lose marks

- L1 Angles measured from the mirror
- L2 “The image is on the mirror”
- L3 Mixing light like paint
- L5 Refraction direction reversed
- L6 “Light bends because it hits the glass”
- L7 Missing arrows and dotted lines

Summary

- Straight lines: reflected, absorbed, transmitted, refracted
- Objects reflect their colour; filters transmit theirs
- Mirror: $i = r$ from the normal; image virtual, behind
- Slower medium: towards the normal

Homework

Light: Unit Review Homework

About 45 minutes

Section A: Achieved

Section B: Merit and Excellence

Use a ruler for every ray. Explain answers in full sentences.